

The Brent-Price Causal Effect of the 2026 Hormuz Disruption

A Synthetic Control Approach — Milestone 3

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Where we left off

Major feedbacks:

- 1 **Stronger justification that the selected donors satisfy SUTVA** — i.e. that no donor is itself contaminated by the treatment.
- 2 **Economic interpretation of why the models behaved as they did** during the treatment period
- 3 **Explore a naive counterfactual** — show what the SCM machinery buys us over simpler baselines.

Re suggested reading **Ham & Miratrix**: it concerns pre-trend identification for *matching / difference-in-differences* designs, not synthetic control on a price series, so its specific machinery does not transfer to our setting

Comment 1 — how we defend donor selection (SUTVA)

Does the asset co-move with Brent through common global factors, yet stay off the oil-supply / disruption path?

Approaches:

- ⇒ **(Main) Economic audit.** (Abadie 2021) Screen every candidate against two contamination channels: the **oil-supply path** (energy complex, petrocurrencies — categorically out) and the **disruption itself** (Hormuz routing, Russia/Ukraine trade — screened donor-by-donor).
- + **Statistical cross-check.** Two tests of whether a donor's *own* series shifts at T_0 — a **permutation mean-shift** test (Chow 1960) and the **HLT** trend-break test (Harvey, Leybourne & Taylor 2009).

Economic audit — Metals & Agriculturals

Metals — Silver, Platinum, Gold

- **Include:** real-rate / safe-haven / dollar channel — a monetary co-movement with Brent, off the oil-supply path.
- **Exclude:** Copper, IronOre (Russia metal supply $\sim 4\text{--}5\%$ world), Palladium (Russia $\approx 40\%$ world supply, spiked Mar-2022).

Agriculturals — Coffee, Sugar, LiveCattle

- **Include:** global trade / demand and the dollar; producers outside the conflict zones (Brazil, India, US).
- **Exclude:** Wheat, Corn (Russia+Ukraine $\approx 30\%$ / 13% world exports), Soybeans (grain cross-elasticity), Cotton (oil $\rightarrow$ polyester substitution).

Economic audit — Equities & FX

Equities — SP500, Nikkei

- **Include:** global growth and the equity risk premium — the demand channel shared with oil; no Russia or Gulf-energy weight.
- **Exclude:** EM_Eq (Russia index weight); energy-heavy DAX / FTSE / TSX-60.

FX — AUD, JPY, CHF, CNY, INR, KRW, ZAR, MXN

- **Include:** safe-haven (JPY, CHF), EM-growth (ZAR, MXN, INR, KRW), China-demand (CNY, AUD) — dollar and growth factors.
- **Exclude:** EUR, DXY (EU energy-crisis / dollar on the invasion), GBP (UK gas); petrocurrencies (CAD, NOK, BRL, ...).

Economic audit — Rates / credit & Volatility

Rates / credit — TLT, HYG

- **Include:** TLT (US duration) and HYG (credit-risk premium) — the discount-rate / risk side of the global factor, not oil-linked.
- **Exclude:** US10Y (Russia: inflation-expectations / flight-to-quality loads it onto the treatment).

Volatility — VIX

- **Include:** the single risk-on / risk-off switch — the fear premium that also lifts oil in stress.
- **Exclude:** none — one volatility proxy, kept deliberately to avoid over-weighting the risk channel.

Net of the audit: 19 of 32 candidates retained (shared pool, both events — 13 dropped for Russia exposure).

Statistical cross-check — shifts at T_0

- **Permutation mean-shift** (Chow 1960): split the donor's own series at T_0 , take the mean gap (after – before), and compare it to thousands of random before/after relabelings — an unusually large gap signals a real shift. Nonparametric, no distributional assumption.
- **HLT trend-break** (Harvey, Leybourne & Taylor 2009): tests whether the donor's own *trend* breaks at T_0 , valid whether the price is stationary or a random walk. Brent-free, so a break implicates the donor itself, not Brent's own T_0 move.

Flagged at T_0 (of 19 retained)	Hormuz 2026	Russia 2022
HLT trend-break	0 / 19	0 / 19
Permutation mean-shift	0 / 19	1 / 19 (CNY)

HLT flags nothing. The lone flag — **CNY** — is the concurrent China zero-COVID reopening, not Russia-treatment, and is kept on the audit's judgment. *Tests inform; the audit decides.*

Comment 2 — why the models behaved as they did

*Why did the five models move (and disagree) the way they did during the event — and does that behaviour **inflate** the premium?*

Approaches:

- 1 Behaviour via donor weights.** Which donors each model leans on to explain Hormuz gap and the Russia divergence?
- 2 Leave-one-out robustness.** Does any single donor drive the gap?
- 3 Signed contamination bias.** Are those donors themselves moved by the treatment (demand destruction)? Decompose each gap into a clean part and a donor-contamination part, and *sign* it.

Behaviour via donor weights



Hormuz. Weight on safe-haven / non-oil donors (Gold, JPY, Sugar) — they track common risk but *not* the chokepoint shock $\Rightarrow$ synthetic undershoots Brent $\Rightarrow$ positive gap.

Russia — divergence is donor-driven. Models lean on *different* donors (Coffee/Platinum vs SP500); Bayesian Ridge degenerates ($w \approx 1.8-2.4$) $\Rightarrow$ dropped from the IQR.

Each column = that model's own importance metric (convex weight / $|\hat{\beta}_j|$ / XGBoost gain), normalised over the 19 donors. Gray = not used.

Leave-one-out robustness

Procedure. Drop one influential donor ($w > 5\%$) at a time, refit, recompute the ATT. Report the **range** (max – min, pp) across the leave-one-donor-out refits — a small range means no single donor drives the result.

Model	Hormuz LOO range (pp)	Russia LOO range (pp)
Convex SCM	8.2	26.8
ASCM	5.7	19.1
Elastic-net	6.7	18.8
XGBoost	6.1	13.8
Bayesian Ridge	10.4	27.3 [†]

Source: `data/validation/inference_loo_{event}_{model}.csv`.

- **Hormuz: tight** (~ 6 – 10 pp on a $\sim 59\%$ effect) — no single donor flips the result; ASCM tightest.
- **Russia: wide** (14 – 27 pp). Bayesian Ridge ([†] range crosses zero, -21.9 to $+5.4$) is LOO-fragile, consistent with its degenerate fit; XGBoost is the most stable.

Signed contamination bias

The critique. If high oil *crowds out demand*, activity-proxy donors are pushed *down*, dragging the synthetic down $\rightarrow$ contamination in the gap. **Which way, and how big?** We sign each linear model's net donor-bias ($-\sum_j w_j \delta_j$) over the full post-window:

Model	Hormuz net bias	Russia net bias
Convex SCM	-1.3% (conservative)	+21.4%
ASCM	-2.1% (conservative)	+12.7%
Elastic-net	-2.4% (conservative)	+20.6%
Bayesian Ridge	-7.6% (conservative)	-7.1%

Linear models only (XGBoost nonlinear). Source: contamination_bias_decomposition.csv.

- **Hormuz: bias negative** for all linear models $\rightarrow$ demand-destruction makes the SCM *under-state* the premium, so the headline **~59%** is a **lower bound**.
- **Russia: bias positive** $\rightarrow$ retained cyclical donors (Coffee, Platinum, SP500) were pushed *down*, dragging the synthetic down and *inflating* the gap $\rightarrow$ Russia is a magnitude check, not the headline.

Comment 3 — what does the SCM buy us over naive baselines?

*If you **ignore the donor pool** and just extrapolate Brent's own past, what counterfactual do you draw — and how far off is the premium?*

Approaches:

- 1 Donor-free baselines.** Build univariate counterfactuals on Brent alone — random walk (flat / drift), linear pre-trend, ARIMA and variants
- 2 Benchmark the premium.** Project each over the treatment window and compare its implied gap to the donor-informed SCM — exposing baselines that freeze or extrapolate blindly.

Donor-free naive baselines

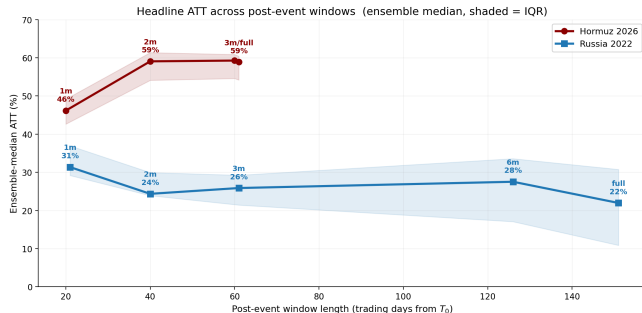
What does the donor-informed SCM buy us over baselines that ignore the donors?



- **Freeze / revert baselines** (flat & AR random walks, mean-reversion) *understate* — Hormuz $\sim 52\%$, Russia $\sim 9\%$.
- **Trend extrapolation** overshoots — linear pre-trend gives Hormuz **73.8%** (overstate); on Russia the extrapolated rally *flips the gap negative* (linear -4.2% , RW+drift -6.9%).
- **SCM** alone tracks the donor-implied path: **58.9%** Hormuz, **22.0%** Russia.

Summary — headline ATT across post-event windows

- 1 **SUTVA** — audit (oil-supply / disruption path), corroborated by the breakpoint test.
- 2 **Model behaviour** — weight on risk donors explains the gap; contamination bias signed (Hormuz conservative); robust to leave-one-out.
- 3 **Naive baselines** — random walks understate, trend; the SCM sits between.



Hormuz builds to **~59%** by month 2–3 and holds (IQR [54, 60]); **Russia** peaks **~31%** (1m) and settles **~22%** (full) — the magnitude check.

Thank you

Questions and discussion

Appendix — in-space placebo (treated-unit inference)

Procedure (Abadie 2010): refit the SCM treating each donor as treated; rank Brent's post/pre RMSPE ratio against the $N=20$ units (Brent + 19 donors). **Permutation** $p = \text{rank} / N$; floor $p = 1/20 = 0.05 \rightarrow$ rank 1 is the strongest signal the 19-donor pool can produce.

Model	Hormuz 2026	Russia 2022
	rank $\rightarrow p$	rank $\rightarrow p$
Convex SCM	1 $\rightarrow$ 0.05	10 $\rightarrow$ 0.50
ASCM	1 $\rightarrow$ 0.05	12 $\rightarrow$ 0.60
Elastic-net	1 $\rightarrow$ 0.05	8 $\rightarrow$ 0.40
XGBoost	1 $\rightarrow$ 0.05	4 $\rightarrow$ 0.20
Bayesian Ridge	1 $\rightarrow$ 0.05	12 $\rightarrow$ 0.60

Source: data/validation/inference_inspace_brent_summary.csv (19-donor shared pool).

- **Hormuz:** Brent ranks 1 for **all five models** $\rightarrow$ its break beats every placebo; strongest signal the pool allows.
- **Russia:** mid-pack (rank 8–12) for the linear models; XGBoost is best at rank 4 ($p=0.20$) but still not significant — consistent with Russia as a magnitude check, not a clean inferential case.

Appendix — walk-forward CV RMSE (model fit)

5-fold expanding-window walk-forward CV (§5a): **val RMSE** = pooled out-of-sample error, **train RMSE** = cross-fold mean. **val/train** > 2 is a heuristic over-fit flag (flagged models are kept — the ensemble takes the median).

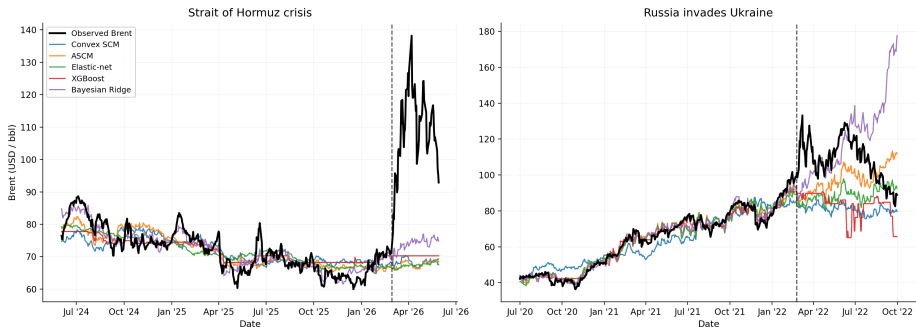
Event	Model	train RMSE	val RMSE	val/train	flag
Russia	Convex SCM	0.113	0.141	1.25	OK
Russia	ASCM	0.046	0.095	2.06	overfit
Russia	Elastic-net	0.051	0.102	1.99	OK
Russia	XGBoost	0.039	0.130	3.31	overfit
Russia	Bayesian Ridge	0.034	0.105	3.09	overfit
Hormuz	Convex SCM	0.060	0.083	1.40	OK
Hormuz	ASCM	0.049	0.075	1.54	OK
Hormuz	Elastic-net	0.052	0.079	1.51	OK
Hormuz	XGBoost	0.045	0.083	1.85	OK
Hormuz	Bayesian Ridge	0.034	0.078	2.28	overfit

RMSE on log-Brent. Source: data/validation/validation_summary.csv. **Russia** 2/5 clean (volatile 2020–22 pre-period); **Hormuz** 4/5 clean — flagged models stay in, the ensemble median is robust.

Appendix — per-model synthetic paths

Observed Brent (black) vs each model's synthetic counterfactual. Pre- T_0 the synthetics *track* Brent (the fit); post- T_0 they diverge — the gap between observed and synthetic is the premium.

Observed Brent vs per-model synthetic counterfactuals (19-donor pool)



Convex SCM & ASCM are the headline pair; Bayesian Ridge's Russia path blows up (degenerate weights, w outside the simplex) — which is why it is down-weighted in the ensemble.

Appendix — interactive figures online

Interactive HTML versions of the EDA figures (zoom, hover, toggle traces) are hosted on GitHub Pages at elviscasco.github.io/Brent_Analysis. The donor panels show the **19-donor audited pool** (VIX retained). Data embedded inline; only the Plotly rendering library loads from CDN.

Figure	URL
Brent + curated events, 2000–2026	plots/eda_brent_events_timeline.html
Donors vs Brent, Hormuz panel	plots/eda_donors_audited_hormuz.html
Donors vs Brent, Russia panel	plots/eda_donors_audited_russia.html
Event-window response heatmap	plots/eda_event_window_response_heatmap.html
Brent + Geopolitical Risk overlay	plots/eda_brent_gpr.html

Source: notebooks/01_EDA.ipynb; outputs regenerated on every notebook run.